ENCS4380 Micromouse Project: M0 Milestone

One-Page Review & Strategic Action Plan

Project: Autonomous Maze-Solving Micromouse

Date: November 1, 2026

Instructor: Dr. Wasel Ghanem

1. Architecture & System Design Choices

- **Mechanical & Chassis:** Custom 3D-printed lightweight chassis modeled in CAD, designed with dedicated mounting slots for the ESP32, motor driver, Li-ion battery pack, and distance sensors to ensure balanced weight distribution.
- **Electronics & Sensing:** ESP32 main controller paired with ToF (Time-of-Flight) sensors for precise wall distance readings, and quadrature encoders coupled with a PID controller to maintain straight-line stability.
- **Firmware & Simulation:** Core maze-solving algorithms—Right-Hand Rule, Floodfill dynamic mapping, and SpeedRun shortest-path execution—validated within the MMS simulation environment prior to physical testing.

2. Lessons Learned & Technical Challenges

- **Timed Turn Discrepancies:** Open-loop time-based rotations proved inaccurate due to motor manufacturing variances and fluctuating battery supply levels.
- **Hardware vs. Simulation Noise:** Physical ToF sensors introduced signal jitter and occasional read timeouts compared to ideal, noise-free simulation data.
- **Volatile Parameter Storage:** Calibration metrics and sensor offset values stored strictly in RAM were reset upon every power cycle.

3. Strategic Action Plan (Our Team Strategy)

- **Closed-Loop Navigation:** Replace timed turns with encoder and IMU feedback combined with active ToF wall realignment after each cell turn.
- **Non-Volatile Flash Storage:** Store all sensor offsets, PID gains, and calibration metrics in flash memory with an automated startup routine.

- **Structured Progress & Workflow:** Maintain strict Git version control with early commits from all team members, alongside real-time task tracking on Trello.